

Homework Assignment 11 (Rewrite)

Connor Mooneyhan

December 5, 2025

7A.17. Suppose μ is a measure, $1 \leq p \leq \infty$, and $f \in \mathcal{L}^p(\mu)$. Prove that for every $\varepsilon > 0$, there exists a simple function $g \in \mathcal{L}^p(\mu)$ such that $\|f - g\|_p < \varepsilon$.

Proof. First consider $p < \infty$. Let $\varepsilon > 0$, and let (g_n) be a sequence of simple functions such that $|g_n| \leq |f|$ and $g_n \rightarrow f$ pointwise. Then $|f - g_n|^p \leq 2^p |f|^p$ and $|f - g_n|^p$ converges to 0 pointwise. Then $2^p |f|^p$ dominates $|f - g_n|^p$ in the sense of Lebesgue's Dominated Convergence Theorem, and we see that it is in $\mathcal{L}^1(\mu)$ because

$$\int |2^p |f|^p| d\mu = 2^p \int |f|^p d\mu = 2^p \|f\|_p^p < \infty,$$

so

$$\lim_{n \rightarrow \infty} \int_X |f - g_n|^p d\mu = \int_X \lim_{n \rightarrow \infty} |f - g_n|^p = \int_X 0 d\mu = 0.$$

We can then choose an N such that $\int_X |f - g_n|^p d\mu < \varepsilon^p$ for all $n \geq N$, so $\|f - g_N\|_p < \varepsilon$ as desired.

When $p = \infty$, we note that (g_n) converges uniformly to f on the set $\{x : |f(x)| \leq \|f\|_\infty\}$, which has measure zero. This allows us to choose an N such that $|f - g_n| < \varepsilon$ for all $n \geq N$. Thus $\|f - g_N\|_\infty < \varepsilon$. $\square$

7A.18. Suppose $1 \leq p < \infty$ and $f \in \mathcal{L}^p(\mathbb{R})$. Prove that for every $\varepsilon > 0$, there exists a step function $g \in \mathcal{L}^p(\mathbb{R})$ such that $\|f - g\|_p < \varepsilon$.

Proof. Fix $\varepsilon > 0$. Using the previous problem, let g be a simple function

$$g = \sum_{i=1}^n a_i \chi_{A_i}$$

such that $\|f - g\|_p < \varepsilon/2$. Then we can find an E_k for each k such that E_k is a finite union of disjoint open intervals and $|E_k \setminus A_k| + |A_k \setminus E_k| < (\frac{\varepsilon}{2|a_k|n})^p$, and so that the E_k are all disjoint. This means

$$\|\chi_{A_k} - \chi_{E_k}\|_p = \|\chi_{A_k} - \chi_{E_k}\|_1^{1/p} < \frac{\varepsilon}{2|a_k|n}.$$

Now we can determine

$$\begin{aligned} \left\| f - \sum_{k=1}^n a_k \chi_{E_k} \right\|_p &\leq \left\| f - \sum_{k=1}^n a_k \chi_{A_k} \right\|_p + \left\| \sum_{k=1}^n a_k \chi_{A_k} - \sum_{k=1}^n a_k \chi_{E_k} \right\|_p \\ &< \frac{\varepsilon}{2} + \sum_{k=1}^n |a_k| \|\chi_{A_k} - \chi_{E_k}\|_p \\ &< \varepsilon \end{aligned}$$

□

7A.19. Suppose $1 \leq p < \infty$ and $f \in \mathcal{L}^p(\mathbb{R})$. Prove that for every $\varepsilon > 0$, there exists a continuous function $g : \mathbb{R} \rightarrow \mathbb{F}$ such that $\|f - g\|_p < \varepsilon$ and the set $\{x \in \mathbb{R} : g(x) \neq 0\}$ is bounded.

Proof. For every $a_1, \dots, a_n, b_1, \dots, b_n, c_1, \dots, c_n \in \mathbb{R}$ and $g_1, \dots, g_n \in \mathcal{L}^1(\mathbb{R})$, we have

$$\begin{aligned} \left\| f - \sum_{k=1}^n a_k g_k \right\|_p &\leq \left\| f - \sum_{k=1}^n a_k \chi_{[b_k, c_k]} \right\|_p + \left\| \sum_{k=1}^n a_k (\chi_{[b_k, c_k]} - g_k) \right\|_p \\ &\leq \left\| f - \sum_{k=1}^n a_k \chi_{[b_k, c_k]} \right\|_p + \sum_{k=1}^n |a_k| \left\| \chi_{[b_k, c_k]} - g_k \right\|_p. \end{aligned}$$

By the previous problem, we can choose $a_1, \dots, a_n, b_1, \dots, b_n, c_1, \dots, c_n \in \mathbb{R}$ so that the term $\left\| f - \sum_{k=1}^n a_k \chi_{[b_k, c_k]} \right\|_p$ is as small as we want. Furthermore, we have previously shown that we can choose $g_1, \dots, g_n$ such that $\left\| \chi_{[b_k, c_k]} - g_k \right\|_1$ is arbitrarily small, and since $0 \leq \chi_{[b_k, c_k]} - g_k \leq 1$ for each k in that construction, we see that

$$\left\| \chi_{[b_k, c_k]} - g_k \right\|_1 \geq \left\| \chi_{[b_k, c_k]} - g_k \right\|_p^p,$$

and thus we can choose $\sum_{k=1}^n |a_k| \left\| \chi_{[b_k, c_k]} - g_k \right\|_p$ as small as we want. We can then finish off the inequalities above as

$$\left\| f - \sum_{k=1}^n a_k g_k \right\|_p < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Define $g = \sum_{k=1}^n a_k g_k$, and we are done. □

5A.1. Suppose (X, S) and (Y, T) are measurable spaces. Prove that if A is a nonempty subset of X and B is a nonempty subset of Y such that $A \times B \in S \otimes T$, then $A \in S$ and $B \in T$.

Proof. We have seen that for any $a \in A$ and $b \in B$, $[A \times B]_a = B$ and $[A \times B]^b = A$ (5.5), and furthermore these cross sections are continuous (5.6), so since $A \times B \in S \otimes T$, we have $A \in S$ and $B \in T$. □

5A.4. Suppose (X, S) and (Y, T) are measurable spaces. Prove that if $f : X \rightarrow \mathbb{R}$ is S -measurable and $g : Y \rightarrow \mathbb{R}$ is T -measurable and $h : X \times Y \rightarrow \mathbb{R}$ is defined by $h(x, y) = f(x)g(y)$, then h is $(S \otimes T)$ -measurable.

Proof. Define $\bar{f} : X \times Y \rightarrow \mathbb{R}$ by $\bar{f}(x, y) = f(x)$ and $\bar{g} : X \times Y \rightarrow \mathbb{R}$ by $\bar{g}(x, y) = g(y)$. Then given measurable $A \in \mathbb{R}$,

$$\begin{aligned} (\bar{f})^{-1}(A) &= \{(x, y) : f(x) \in A\} \\ &= f^{-1}(A) \times Y. \end{aligned}$$

Since f is measurable, $f^{-1}(A)$ is measurable, and thus $f^{-1}(A) \times Y \in S \otimes T$ since it is a measurable rectangle. By the same reasoning, $(\bar{g})^{-1}(A) = X \times g^{-1}(A) \in S \otimes T$.

Since $h(x, y) = f(x)g(y) = \bar{f}(x, y)\bar{g}(x, y)$, h is now simply a product of $(S \otimes T)$ -measurable functions, and so it is itself $(S \otimes T)$ -measurable. □

5A.8. Suppose μ is a measure on a measurable space (X, S) . Prove that the following are equivalent.

- (a) The measure μ is σ -finite.
- (b) There exists an increasing sequence $X_1 \subset X_2 \subset \cdots$ of sets in S such that $X = \bigcup_{k=1}^{\infty} X_k$ and $\mu(X_k) < \infty$ for every $k \in \mathbb{Z}^+$.
- (c) There exists a disjoint sequence $X_1, X_2, X_3, \dots$ of sets in S such that $X = \bigcup_{k=1}^{\infty} X_k$ and $\mu(X_k) < \infty$ for every $k \in \mathbb{Z}^+$.

Proof. (a $\implies$ b) Let $X = \bigcup_1^{\infty} A_i$ where each $A_i \in S$ and $\mu(A_i) < \infty$. Then define $X_k = \bigcup_1^k A_i$. For all k , we have $X_k \subset X_{k+1}$, $\mu(X_k) \leq \sum_1^k \mu(A_i) < \infty$, and $\bigcup_1^{\infty} X_k = \bigcup_1^{\infty} A_i = X$, as desired.

(b $\implies$ c) Let $A_1 \subset A_2 \subset \cdots$ be a sequence of sets in S such that $X = \bigcup_1^{\infty} A_i$. Set $A_0 = \emptyset$ (which changes none of the properties mentioned above), and define $X_k = A_k \setminus A_{k-1}$ for all $k \geq 1$. Then we have $\bigcup_1^{\infty} X_k = X$, and for all k , $\mu(X_k) < \mu(A_k) < \infty$. Also, the sequence is disjoint as desired.

(c $\implies$ a) Let $X_1, X_2, \dots$ be as specified in (c). Then $X_1, X_2, \dots$ is itself a countable collection of sets of finite measure whose union is X , so μ is σ -finite. $\square$